

# RECURRENT NEURAL NETWORKS

Pain points

## RNN

Can we please briefly go through the data preparation for RNN, what is a sequence size how is it used, why we use SGD even though you said not to use it and can we use sth else than SGD or it won't work?

Also why are we adding the 56days before the test set to the test set?

## RNN

Can we please briefly go through the data preparation for RNN, what is a sequence size how is it used,

```
seq_length = 56
train_ds = tf.keras.utils.timeseries_dataset_from_array(
    np.asarray(train_set).astype('float32'),
    targets = train_set[["rail"]][seq_length:],
    sequence_length = seq_length,
    batch_size = 32,
    shuffle = True,
    seed = 42
)
```

[5, 7, 2, 9, 6, 11, 4]

seq\_length = 3

X  
features

|   |   |    |
|---|---|----|
| 5 | 7 | 2  |
| 7 | 2 | 9  |
| 2 | 9 | 6  |
| 9 | 6 | 11 |

y  
target

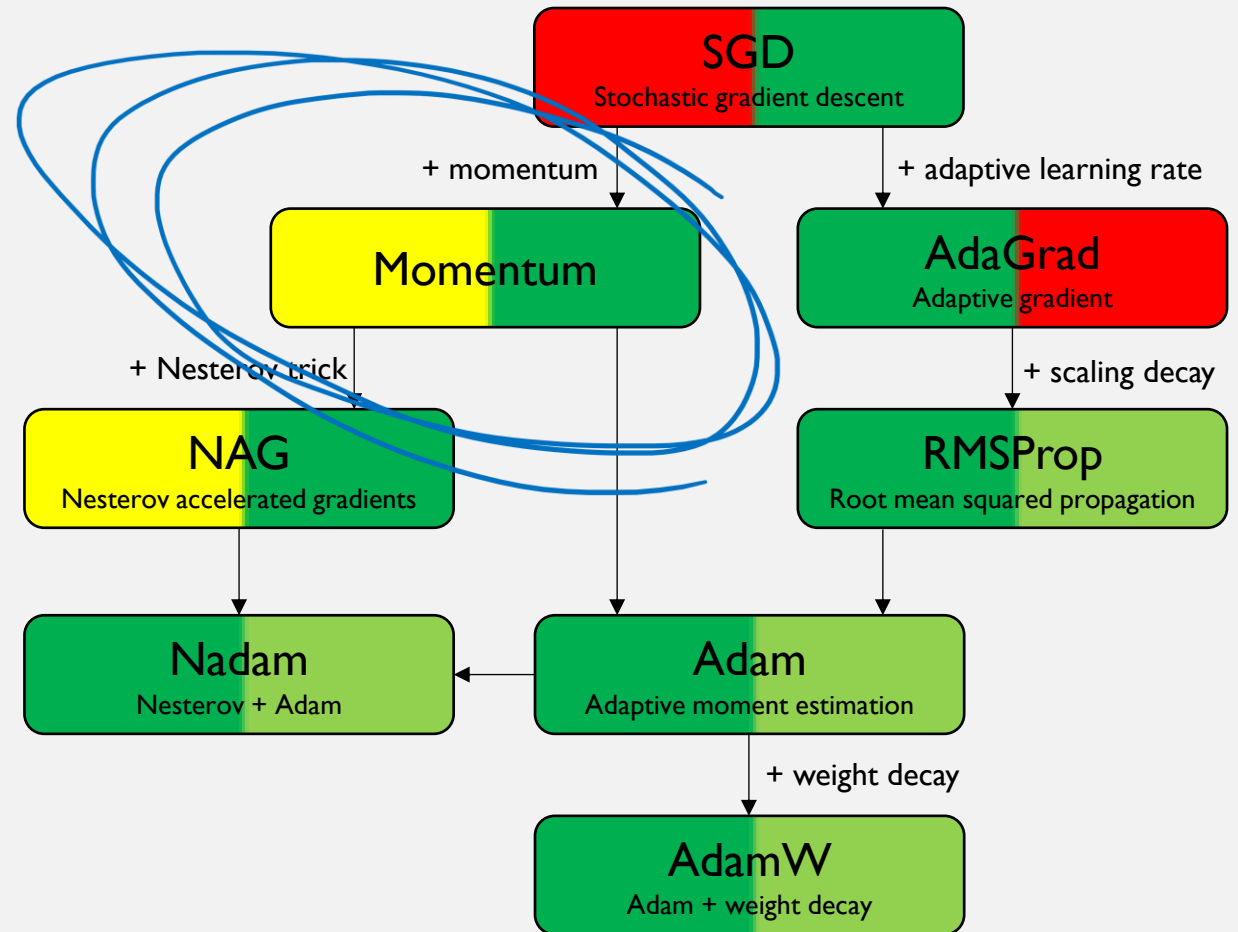
↓

|    |
|----|
| 9  |
| 6  |
| 11 |
| 4  |

why

we use SGD even though you said  
not to use it and can we use sth else  
than SGD or it won't work?

```
optimizer = tf.keras.optimizers.SGD(learning_rate = 0.02, momentum = 0.9)
```



Also why are we adding the 56days  
before the test set to the test set?

```
test_set_with_prev_56_days = df_rnn["2019-01-04":"2019-06"]
```

for visualization purposes

/ to predict the 1st  
day in the test set